

PRERAK PANDEY

Chicago, IL | prerak.pandey7@gmail.com | 331-254-8181 | github.com/prerak7-dev/AegisMotionSandbox | Open to Amsterdam relocation

ONLINE SERVICES / BACKEND ENGINEER

PROFESSIONAL SUMMARY

Software engineer with 3+ years building full-stack web applications, CMS integrations, cloud-ready deployments, and cross-functional delivery workflows, now focused on online game services and developer-facing game technology. Portfolio work includes AegisMotionSandbox: a Java/Spring Boot + Redis/Kafka animation processing pipeline paired with an Unreal Engine C++ plugin for data-driven procedural animation. Practical strength in maintainable service architecture, REST/GraphQL integrations, asynchronous processing, CI/CD-aware development, debugging, and clear technical communication between engineering and content-focused teams.

CORE SKILLS

Backend & Services: Java, Spring Boot, C#, .NET, Node.js, REST APIs, GraphQL, Kafka, Redis, SQL/PostgreSQL, MySQL, Sitecore Headless/JSS services

Cloud & Delivery: AWS S3, Cognito, IAM, Lambda; Azure; Docker; CI/CD; Git; deployment support; CDN/cache workflows; logging and release diagnostics

Game / Tools Engineering: Unreal Engine C++, Animation Blueprints, custom AnimGraph nodes, editor tooling, data assets, procedural animation, mocap-to-curves workflows

Frontend: Next.js, React, Angular, TypeScript, JavaScript, headless CMS rendering

Collaboration: Requirement translation, estimation, stakeholder communication, technical documentation, cross-disciplinary problem solving

PROFESSIONAL EXPERIENCE

Web Application Developer | Americaneagle.com — Des Plaines, IL | Jul 2021 – Apr 2024

- Developed and maintained enterprise Sitecore and headless web applications across insurance, nonprofit, medical/scientific, and data-sector clients, working across frontend, backend, CMS integrations, and deployment support.
- Contributed to Westfield Insurance Sitecore 10.3 upgrade using Next.js, Sitecore Headless/JSS services, GraphQL layout and dictionary APIs, Sitecore CDP/XMCloud integration, and personalized content authoring workflows.
- Built a custom Sitecore extension for dynamic CDN cache purging on content publish for United Methodist Church, improving release reliability for live content updates and reducing manual cache intervention.
- Set up Sitecore Publishing Service and Publishing Service Module in a Sitecore 10.2 environment; supported CMS maintenance, analytics/deployment reporting, and CI/CD operations across Sitecore 9/10 solutions.
- Took lead responsibility for estimation, requirement translation, and delivery coordination on long-running client work, bridging business needs with implementation details for developers and stakeholders.
- Worked in Git-based team workflows with recurring debugging, release support, data integrations, and production-minded communication across distributed project teams.

Student Data Analyst | Center for Student Success, Ramapo College of New Jersey — Mahwah, NJ | Dec 2017 – May 2021

- Analyzed academic and non-academic student data across departments to support advising prioritization, strategic planning, and resource allocation decisions.
- Automated recurring data processing and reporting workflows with advanced spreadsheets, macros, and scripting to reduce manual reporting overhead.
- Built student-success predictor concepts from performance and feedback signals, helping advisors identify students who needed earlier support.
- Presented analysis to academic stakeholders in clear, decision-focused language for non-technical and technical audiences.

PORTFOLIO ENGINEERING PROJECT

AegisMotionSandbox — Online Animation Processing Pipeline + Unreal Engine Plugin | Independent Project — GitHub | 2024 – Present

- Built an end-to-end game-technology portfolio project that connects a Java/Spring Boot service pipeline with an Unreal Engine C++ plugin for data-driven procedural animation.
- Designed a multi-service animation pipeline with ingestion and processing services, using Kafka for asynchronous job messaging, Redis for job state/cache coordination, and JSON overlays as the contract into Unreal tooling.
- Modeled service boundaries around online-systems concerns: non-blocking job processing, job isolation, retry-friendly workflow stages, structured logging, validation, and artifact handoff into engine tools.
- Developed the Aegis Motion Unreal plugin with a procedural action data asset, runtime action component, custom AnimGraph driver, curve-driven bone transforms, debug scrub controls, and viewport visualization for authoring.
- Implemented workflows for BVH/FBX/JSON animation sources to populate editable procedural curves that designers can tune through data assets rather than hard-coded animation logic.
- Separated runtime plugin logic from editor tooling, keeping animation behavior declarative and exposing debug controls for diagnosing blends, curve values, bone-chain transforms, and runtime state.

EDUCATION

Ramapo College of New Jersey — Bachelor of Science, Computer Science | Jun 2021

SELECTED LINKS

AegisMotionSandbox: github.com/prerak7-dev/AegisMotionSandbox | Client deployments: umc.org, asq.org, aans.org, westfieldinsurance.com